Low-degree all-width learning bounds for shallow Bayesian neural networks

ChatGPT-6 Astra Kaarel Hänni Dmitry Vaintrob*

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Abstract

For a tanh network with one hidden layer, we prove an all-width Gaussian Bayesian learning bound essentially linear in the weighted squared norm of a small teacher. The result includes the raw Gaussian prior, the prior conditioned on an absolute margin everywhere, and a specified robust prior. Training requires signed outputs above a fixed positive margin. The sample bound is independent of learner width and holds at every mean-field or stronger Gaussian scale. With hidden coefficients at most the teacher width and bounded readout coefficients, the width dependence is quadratic. Allowing the readout coefficients also to reach the width gives a quartic sufficient bound. The proof represents tanh ridges as mixtures of halfspaces, compresses their votes, and controls posterior readout energy. Independent AI-agent review checked the argument; it is formalized in Lean for the main learning statements, with scope recorded in the repository.

1 Result and scope

For one hidden layer, the mean-field-and-stronger Gaussian learner admits a direct finite-class argument. Its sample bound is linear, up to logarithms, in a normalized target-cylinder cost. No deep-network compression is needed.

Let $d, C \geq 1$, put $\bar{x} = (x, 1)$, and write the learner as $z_w(x) = \sum_{r=1}^{N} v_r \tanh(W_r \cdot \bar{x})$. For a teacher

$$z_*(x) = \sum_{j=1}^{C} a_j \tanh(b_j \cdot \bar{x}), \qquad f(x) z_*(x) > \tfrac{1}{2} \quad (x \in \{-1, 1\}^d),$$

write $A_1 = \sum_j |a_j|$, $A_2^2 = \sum_j a_j^2$, and $U = d \min\{\max_j \|b_j\|_2^2, (2/C) \sum_j \|b_j\|_2^2\}$. The learner has arbitrary width $N \geq C$, independent Gaussian entries of variance t^2/d in the first layer and t^2/N^2 in the readout, where $0 < t \leq 1$.

Any of the following priors is allowed:

1. The Gaussian prior conditioned on $|z(x)| > 1/2$ at every Boolean input.
2. A robustness-restricted prior, using one hidden layer, radius $c/\sqrt{1 + E_v}$, $E_v = N\|v\|_2^2$, clipped-score tolerance $1/16$, and optional additional global-margin conditioning.
3. The raw Gaussian prior, with no initial margin or robustness restriction.

*Authors are listed in alphabetical order.

Here the shallow robustness event means the following. Put $T(u) = \max\{-1/2, \min\{1/2, u\}\}$, add disturbances by $z_w(x;\eta) = v^\top[\tanh(W\bar{x}) + \eta^1] + \eta^{\text{out}}$, and let $\|\eta\|_*^2 = \|\eta^1\|_2^2/N + |\eta^{\text{out}}|^2$. The event requires

$$|T(z_w(x;\eta)) - T(z_w(x))| \le 1/16 \quad \text{for every } x \in \{-1,1\}^d, \quad \|\eta\|_* \le c/\sqrt{1 + N\|v\|_2^2}.$$

The calibration $c > 0$ is fixed before training.

Fix an arbitrary input distribution D on the Boolean cube and draw the inputs independently from D. In all three cases, training conditions on $f(X_i)z(X_i) > 1/2$. Test error means $f(X)z(X) \le 1/4$, which bounds ordinary sign error. With global-margin conditioning, the two errors coincide; this equality is not asserted for the raw or robust-only priors.

Let $s = 3$ in cases 1 and 3, and $s = 2c + 3$ in case 2. Set

$$\rho = \frac{1}{4[1 + sA_1(d+1)/\sqrt{d}]}, \qquad D_* = U + 2s^2CA_2^2 + (d+2)\log\frac{e\sqrt{2\pi}}{2\rho}.$$

Then $D_* > 1$. For $0 < \varepsilon, \delta < 1$, define

$$B^2 = 8\left[D_* + \log\frac{4}{\varepsilon}\right], \qquad m = \left\lceil 512B^2 \log\frac{24}{\varepsilon}\right\rceil, \qquad h_d = (d+1)(1 + d\log 2).$$

A sufficient sample size is

$$\boxed{n \ge \frac{16}{\varepsilon}\left[mh_d + \log\frac{1}{\delta}\right].}$$

With probability at least $1 - \delta$, the posterior-averaged test-margin error is at most ε, simultaneously for all $N \ge C$ and $0 < t \le 1$.

Thus the bound is

$$O\left(\frac{d^2[D_* + \log(1/\varepsilon)]\log(1/\varepsilon) + \log(1/\delta)}{\varepsilon}\right).$$

A sharper alternative, proved below by a relative growth-function argument, is obtained by setting

$$T = \frac{64}{\varepsilon}\left[m(d+1) + \log\frac{4(m+2)}{\delta}\right]$$

and taking $n \ge \lceil 4T\log(8eT)\rceil$. Up to logarithms, this is $\widetilde{O}((dD_* + \log(1/\delta))/\varepsilon)$, improving the dimension factor from d^2 to d.

The displayed constants are deliberately loose; the useful improvement is in the powers.

In terms of the teacher's weighted squared norm

$$\kappa_C = \frac{d}{C}\sum_j \|b_j\|_2^2 + C\sum_j a_j^2,$$

one has $D_* \le 2s^2\kappa_C + (d+2)\log(e\sqrt{2\pi}/(2\rho))$. Thus the sample bound is essentially **linear in the teacher's weighted squared norm**.

1.1 Width consequences

If first-layer entries satisfy $|b_{ji}| \le W$ and readout entries satisfy $|a_j| \le V$, then

$$D_* \le d(d+1)W^2 + 2s^2C^2V^2 + (d+2)\log\Big(2e\sqrt{2\pi}[1 + sCV(d+1)/\sqrt{d}]\Big).$$

- For fixed V, with $W \le C$, this gives a genuinely quadratic dependence on teacher width C. The first, finite-cube bound is literally $O(C^2)$ when the other parameters are fixed: its additive $\log C$ target-cost term is absorbed into C^2. The dimension-sharpened bound has an additional logarithmic factor in C. Both statements hold even for the raw Gaussian prior, without an initial margin or robustness restriction.
- Under the original bounded-weight-complexity convention $W, V \le C$, it gives $D_* = O(d^2C^2 + s^2C^4 + d\log(dCs))$, and hence quartic dependence on C. This proof does **not** establish quadratic dependence for that full class.
- Width alone cannot be substituted for the theorem's teacher resources without controlling coefficient magnitudes or another implementation cost.

2 Proof

2.1 1. A target cylinder at every width

Partition the N hidden neurons into C balanced groups with positive sizes k_j. Then $N/k_j \le 2C$ and $k_j/N \le 2/C$. Give each member of group j first-layer row b_j and readout coefficient sa_j/k_j. This clone has output sz_*.

The Gaussian coordinates are $\sqrt{d}\,W_{ri}$ and Nv_r, all independent $N(0, t^2)$. Require every coordinate to lie within ρt of its clone center. On this cylinder,

$$|z_w(x) - sz_*(x)| \le \rho + sA_1\rho(d+1)/\sqrt{d} = \tfrac{1}{4}$$

simultaneously on the entire Boolean cube. The first term is the readout perturbation, using bounded activations; the second is the changed hidden activations weighted by the cloned readout.

In cases 1 and 3 this gives $f(x)z_w(x) > 5/4$. In case 2 it gives $f(x)z_w(x) > c + 5/4$.

For a shallow network, a simultaneous hidden-and-output disturbance obeys

$$|z_w(x;\eta) - z_w(x)| = |v \cdot \eta^1 + \eta^{\text{out}}| \le \sqrt{1 + E_v}\,\|\eta\|_*.$$

Consequently the calibrated ball changes any output by at most c. Every cylinder network in case 2 remains above signed margin $5/4$ throughout that ball. Its clipped score is constant there, so it lies in the robust support. (In fact, for $c \le 1/16$, every shallow network lies in that support, because clipping is 1-Lipschitz.)

There are $N(d+2)$ independent constrained coordinates. Their squared centers sum to at most

$$NU + s^2N^2\sum_j a_j^2/k_j \le N[U + 2s^2CA_2^2].$$

For $Z \sim N(0, t^2)$, $0 < \rho \le 1/4$,

$$\Pr(|Z - u| \le \rho t) \ge \frac{2\rho}{e\sqrt{2\pi}}e^{-u^2/t^2}.$$

Multiplying, and using $t \le 1$, proves that the cylinder has raw mass at least e^{-ND_*/t^2}. It lies in every correctly labeled posterior-conditioning event. In case 3 that event consists only of the training-margin constraints; no initial support restriction is used.

2.2 2. A width-independent readout-energy tail

Under the raw Gaussian, NE_v/t^2 has the chi-square distribution with N degrees of freedom. For $R \ge 8$, its exponential moment gives

$$\Pr(E_v > R) \le e^{-NR/(8t^2)}.$$

Dividing by the posterior denominator from step 1, and taking $R = B^2$, shows that the posterior mass of $E_v > B^2$ is at most $\varepsilon/4$, on every sample. Every retained network has

$$\sum_{r=1}^{N} |v_r| \le \sqrt{N}\,\|v\|_2 \le B.$$

Let Q' be the posterior conditioned on this readout-energy event.

2.3 3. A bounded-variation shallow network is a mixture of halfspaces

Let T have continuous distribution function $\Pr(T \le u) = (1 + \tanh u)/2$. Then

$$\tanh(u) = \mathbb{E}_T \operatorname{sign}(u - T).$$

Thus each ridge $\tanh(b \cdot \bar{x})$ is an average of Boolean affine halfspaces on the cube. A sign change of a halfspace is another halfspace pattern; zero-probability boundary ties can be ignored because the input set is finite.

Every retained output z therefore satisfies

$$z(x) = B\,\mathbb{E}H(x)$$

for a distribution on Boolean affine-halfspace patterns H. Assign mass $|v_r|/B$ to the corresponding signed ridge mixture. Fill any remaining mass with equal probabilities on the two constant halfspaces.

The number M_d of distinct affine-halfspace patterns on the cube is at most

$$M_d \le (e2^d)^{d+1}, \qquad \log M_d \le h_d,$$

by the VC dimension $d + 1$ and Sauer's lemma.

Independently sample m halfspaces from the representation of z and put

$$h(x) = \frac{B}{m} \sum_{j=1}^{m} H_j(x).$$

For every fixed input x, Hoeffding's inequality gives

$$\Pr(|h(x) - z(x)| > 1/16) \le 2e^{-m/(512B^2)} \le \alpha := \varepsilon/12.$$

The compressed outputs lie in one fixed finite family $\mathcal{H}$, with $\log|\mathcal{H}| \le mh_d$. Its sampling distribution may depend on z, but does not depend on the input being tested. Averaging these distributions over $w \sim Q'$ gives a common law Γ on $\mathcal{H}$.

2.4 4. Generalization of the compressed mixture

Use the ramp loss

$$\psi(u) = \begin{cases} 1, & u \le 5/16, \\ (7 - 16u)/2, & 5/16 < u < 7/16, \\ 0, & u \ge 7/16. \end{cases}$$

On a training point, a successful approximation has signed compressed output strictly above $7/16$, because every original posterior draw has signed output above $1/2$. Therefore

$$\mathbb{E}_{h\sim\Gamma}\widehat{R}_{\psi}(h) \le \alpha.$$

At a test input where $fz_w \le 1/4$, successful approximation gives $fh \le 5/16$, hence unit ramp loss. Thus

$$R_{1/4}(Q') \le \mathbb{E}_{h\sim\Gamma}R_{\psi}(h) + \alpha.$$

For a finite family of bounded losses, the relative Chernoff inequality and a union bound imply, with probability at least $1 - \delta$,

$$R_{\psi}(h) \le 2\widehat{R}_{\psi}(h) + \frac{8[\log|\mathcal{H}| + \log(1/\delta)]}{n} \quad \text{for every } h \in \mathcal{H}.$$

This event is independent of N, t, and holds for all mixture laws on $\mathcal{H}$, including the data-dependent law Γ. Hence

$$R_{1/4}(Q) \le \varepsilon/4 + 3\alpha + \frac{8[mh_d + \log(1/\delta)]}{n} \le \varepsilon.$$

This proves the simultaneous theorem.

3 Checks and limitations

- The proof controls the posterior average, so tied minimizers and posterior mixtures cause no problem. A correct-margin posterior confidence threshold $q < 1$ follows by using risk tolerance $(1 - q)\varepsilon$.
- The arbitrary first-layer weights of posterior draws do not enter the compressed family. Only the halfspace patterns matter.
- The raw-prior version still uses hard training-margin conditioning. It does not assert the same result for Gaussian regression, logistic likelihoods, or conditioning only on training signs without a positive margin.
- The theorem works with the existing Gaussian mean-field and stronger scalings: $t = N^{(1-\gamma)/2}$ gives every $\gamma \ge 1$. It also covers every sufficiently strong finite-width scale of the original strong-prior parametrization, by setting $t = \sqrt{N}\sigma \le 1$.
- The first bound uses simple finite-cube counting. The additional argument below improves the dimension factor to d, up to logarithms, by counting the sample patterns of voted halfspaces.
- A Boolean circuit with a single output gate and only one Boolean stage is a small target class regardless of the number of unused wires. The meaningful shallow comparator here is a real-valued tanh hidden layer and linear readout implementing a Boolean function.
- This is an independent derivation, not a claimed optimization of the old degree-510/10974 compression formulas.

4 Sharpening the dimension factor by counting sample patterns

The preceding proof counted halfspaces on the entire cube. Counting only their patterns on a sample improves the dimension factor from d^2 to d, up to logarithms. Here is a self-contained argument with deliberately generous constants.

4.1 A relative growth-function lemma

Let $\mathcal{A}$ be a finite class of measurable sets and let $\Pi_{\mathcal{A}}(r)$ be its maximum number of distinct membership patterns on r points. For an iid sample of size n, with probability at least $1-\delta$,

$$P(A) \le 2P_n(A) + \frac{32}{n}\log\frac{4\Pi_{\mathcal{A}}(2n)}{\delta} \quad \text{for all } A \in \mathcal{A}.$$

To prove this, let $u > 0$ with $nu \ge 32\log 2$, and suppose some set has $p = P(A) > 2P_n(A) + u$. Conditional on the first sample and a selected witness A, an independent ghost sample satisfies $P'_n(A) \ge 3p/4$ with probability at least $1/2$: the lower Chernoff tail is at most $e^{-np/32}$. Hence the probability of a witness is at most twice the probability that

$$P'_n(A) > \tfrac{3}{2}P_n(A) + \tfrac{3}{4}u$$

for some A.

Pool the two samples and split their $2n$ positions uniformly into two groups of size n. A fixed membership pattern with k positive positions has a hypergeometric number K of positives in the first group, with mean $k/2$. The displayed event implies

$$K < \tfrac{2}{5}k - \tfrac{3}{10}nu,$$

so the sampling-without-replacement form of Hoeffding's inequality gives, for $k > 0$,

$$\Pr\left(K < \tfrac{2}{5}k - \tfrac{3}{10}nu\right) \le \exp\left[-\frac{2(k/10 + 3nu/10)^2}{k}\right] \le e^{-6nu/25}.$$

For the last inequality, expand the exponent and use $k/50 + 9(nu)^2/(50k) \ge 3nu/25$. The case $k = 0$ is impossible. A union bound over at most $\Pi_{\mathcal{A}}(2n)$ patterns therefore bounds the witness probability by $2\Pi_{\mathcal{A}}(2n)e^{-6nu/25}$. Taking $u = (32/n)\log[4\Pi_{\mathcal{A}}(2n)/\delta]$ proves the lemma.

4.2 Applying the lemma to the compressed ramp losses

Work on labeled examples (x, y). Consider all sets

$$\{(x, y) : \psi(yh(x)) > q\}, \qquad h \in \mathcal{H}, \quad 0 \le q \le 1.$$

This is a finite set class: each compressed output is a vote of m Boolean halfspaces and has at most $m+1$ possible output values, so varying q gives at most $m+2$ membership patterns for a fixed vote.

On any r input points, one affine halfspace has at most $(er)^{d+1}$ patterns. Choosing m such patterns gives at most $(er)^{m(d+1)}$ voted outputs. Multiplication by the fixed labels of the sample does not increase this number. Consequently the loss-level-set growth function satisfies

$$\Pi(r) \le (m+2)(er)^{m(d+1)}.$$

Apply the relative growth-function lemma to all these level sets and integrate in q. The layer-cake identity for a bounded nonnegative loss gives, simultaneously for every $h \in \mathcal{H}$,

$$R_\psi(h) \le 2\widehat{R}_\psi(h) + \frac{32}{n}\left[m(d+1)\log(2en) + \log\frac{4(m+2)}{\delta}\right].$$

The same averaging and readout-tail argument as above therefore gives posterior risk at most ε whenever

$$\boxed{n \ge \frac{64}{\varepsilon}\left[m(d+1)\log(2en) + \log\frac{4(m+2)}{\delta}\right].}$$

For a fully explicit sufficient bound, put

$$T = \frac{64}{\varepsilon}\left[m(d+1) + \log\frac{4(m+2)}{\delta}\right].$$

Then it suffices to take

$$\boxed{n \ge \lceil 4T\log(8eT)\rceil\,.}$$

Indeed, write $A = 64m(d+1)/\varepsilon$ and $B_0 = 64\log[4(m+2)/\delta]/\varepsilon$, so $T = A + B_0$. At $n_0 = 4T\log(8eT)$, the quantity $A\log(2en_0) + B_0$ is at most $T[\log(8eT) + \log\log(8eT) + 1] \le n_0$. The difference $n - A\log(2en) - B_0$ is increasing for $n \ge n_0 > A$.

Up to logarithmic factors, this yields the sharper sample bound

$$\boxed{n = \widetilde{O}\Big(\frac{dD_* + \log(1/\delta)}{\varepsilon}\Big).}$$

The exact formulas above include the logarithms in ε, D_*, d, and confidence. The sample event remains independent of learner width and Gaussian scale.

4.3 References for the elementary ingredients

- W. Hoeffding, "Probability inequalities for sums of bounded random variables," *Journal of the American Statistical Association* **58** (1963), 13–30. The sampling-without-replacement inequality is included in Section 6.
- N. Sauer, "On the density of families of sets," *Journal of Combinatorial Theory, Series A* **13** (1972), 145–147.

The relative growth-function statement used here was proved above, rather than being assumed as an external theorem.

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